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Vermont AI Guidance for Education

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Resources & Quick Start Guides

This guidance document is comprehensive by necessity because AI in education raises important questions about pedagogy, safety, equity, and implementation. However, we recognize that most readers need specific, actionable information for their role.

Quick Start Guides by Role

- [Elementary Educators](#)
- [Middle School Educators](#)
- [High School Educators](#)
- [IT & School Leaders](#)

Frequently Asked Questions

- [What do you mean by AI?](#)
- [What if I don't know anything about AI?](#)
- [How can I access training and professional development?](#)
- [Which AI tools can I use in my classroom?](#)
- [What if my students use AI inappropriately or without permission?](#)
- [How can I minimize academic honesty violations related to student use of AI?](#)

Essential Resources

- [How to Work with AI for Educators \(Prompting Guide\)](#)
- [Glossary of Terms](#)

Part 1: Foundations

A Note on Terminology

This guide uses the term "AI" (artificial intelligence) throughout. Artificial intelligence is a broad classification. This guide most often refers to generative AI and the agentic systems built on top of this technology. See the [Glossary](#) for complete definitions.

Introduction & Vision

The tight relationships between school staff and students, and close communities that exist in Vermont Schools give us a unique advantage in the age of Artificial Intelligence: we can implement these powerful tools while maintaining the personal connections that define Vermont education. As AI impacts society, Vermont can model a human-centered approach, one that uses AI to enhance rather than replace the relationships between educators, students, and communities. A human-centered approach is necessary not just to enhance learning, but to protect the cognitive and social development of students from the risks of over-reliance and automated thinking.

Navigating this transition requires an intentional balance, one that honors the real risks inherent in adopting these tools by proactively addressing the potential for AI to disrupt student well-being or academic integrity. This balance is maintained through the deliberate guardrails of human agency, where experienced educators ensure technology is used as a tool to support the instructional core. By prioritizing student learning outcomes and the cultivation of content knowledge expertise, schools ensure that technological advancement never supersedes the development of human intellect.

Vermont schools have long championed personalized learning through personalized learning plans (PLP), proficiency-based learning and project-based learning. AI has the potential to amplify these strengths in three powerful ways.

Personalizing Learning at Scale - AI technology can adapt to the unique interests and needs of each learner. Tight integrations of AI with curricular materials can provide tutoring support, freeing teachers to work individually with students. AI also assists in providing accommodations for students with special needs by taking notes, providing support for students with visual or auditory impairments, and assisting students with speech impairments. Additionally, while adaptive learning systems can adjust the pace of instruction, AI empowers students in upper grade levels to leverage these tools within project-based learning to navigate their own individualized learning pathways.

Empowering Educators - AI potentially gives teachers a capable assistant tool that can simultaneously tutor multiple students, differentiate instruction, and handle routine tasks. Teachers gain the power to strategically choose what to delegate to AI and what to handle personally - much like a flipped classroom model but with far more flexibility. While AI handles skill practice and supports content delivery, teachers can focus on the

high-impact moments: guiding discussions, providing emotional support, sparking creativity, and helping students make meaningful connections. AI technology has the potential to adapt to each teacher's approach; however, it's up to us as educators to harness it effectively.

Transforming Student Engagement - AI enables new learning experiences such as immersive simulations, intelligent project partners, real-time language translation for multilingual families, and dynamic activities that respond to students' interests. Students can collaborate with AI to create, explore, and solve problems in ways that prepare them for their future.

Why This Guidance Matters for Vermont

This guidance reflects Vermont's commitment to giving all students roots and wings by grounding students in essential knowledge while preparing them to thrive in a rapidly changing world. In a world where AI can generate text, solve equations, and retrieve information instantly, some have asked whether students still need to learn content deeply. The answer is an emphatic yes.

Students cannot think critically about information they don't understand. They cannot evaluate AI-generated outputs without the background knowledge to recognize errors, biases, or gaps. They cannot ask good questions of AI tools without foundational understanding of the subject matter. Far from making content knowledge obsolete, AI makes it more essential: the students best positioned to use these tools effectively are those with strong foundations in reading, writing, mathematics, and disciplinary knowledge.

The transferable skills defined in the Education Quality Standards and integrated throughout Vermont's Portrait of a Graduate relate directly to the skills students need to be successful in a world transformed by AI. The World Economic Forum's 2025 Future of Jobs Report shows that employers increasingly value creative thinking, analytical reasoning, resilience, and curiosity alongside technological literacy. These are not replacements for traditional academic skills, rather they are built upon them. AI replaces routine tasks, not the human capacity for judgment, problem-solving, and meaning-making that strong education develops.

We must confront two false narratives: that AI will solve all of education's challenges, and that AI will pose only risks to avoid. The greater risk lies in doing nothing, leaving students without the skills to navigate an AI-integrated world, and leaving educators without clear guidance on responsible use.

Students need strong content knowledge now more than ever. To be discerning consumers and ethical users of AI, they must have the background knowledge to evaluate AI outputs, the critical thinking skills to question what they see, and the creativity to apply AI as a tool rather than a crutch. AI literacy is not a replacement for foundational learning; it depends on it.

A Note on Using This Guidance

Vermont's approach to AI should be intentional and problem focused. This guidance explores multiple possibilities. Each school should identify its unique challenges, then determine which AI tools, if any, best address those needs. Technology should serve our educational mission, not define it.

Key Concerns for AI in Education

These opportunities don't come without concerns. While the concerns are highlighted here, they are explored in detail later in this document, along with strategies to mitigate each of these concerns.

- **Academic integrity considerations**
Students may use AI to complete assignments without permission, potentially bypassing the learning objectives and violating academic honesty policies. [Learn how to address this concern.](#)
- **Cognitive offloading**
Over-reliance on AI for tasks can prevent students from developing essential skills through productive struggle and practice. [Learn how to find the right balance.](#)
- **Data privacy and security**
Student data and intellectual property must be protected when using AI systems to comply with FERPA, COPPA, and potentially HIPAA requirements. [Learn how to protect student data.](#)
- **Bias and fairness**
AI systems can perpetuate stereotypes and biases through their training data, potentially leading to unfair or inaccurate outputs. [Learn how to mitigate bias.](#)
- **Student well-being and safety**
AI can be misused to create harmful content. Deepfakes and non-consensual images threaten student safety. [Learn how to protect students.](#)
- **Mental health considerations**
Students may develop unhealthy relationships with AI companions, with 72% of teens having used AI companions at least once. (Common Sense Media, 2025) [Learn how to support healthy AI use.](#)
- **Equitable access**
Not all students have equal access to AI tools and technology, which potentially widens achievement gaps. [Learn how to support equitable access.](#)
- **Ethical considerations**
AI raises concerns about intellectual property, environmental impact, and job displacement that students need to address. [Learn how to navigate these issues with your students.](#)

Part 2: Educational Uses of AI

Developmental Appropriateness of AI

Schools and teachers should feel empowered to determine when student AI usage is inappropriate; for example, AI chatbots are not developmentally appropriate for all students. Developmental appropriateness depends on student maturity as well as AI literacy. The example of a literacy timeline presented here is a modified and condensed version based on North Carolina's AI literacy timeline (North Carolina Department of Public Instruction, 2024).

We can view students' developmental process through three stages:

1. **AI Awareness:** The ability to recognize when AI is present in daily life and understand its basic role as technology trained on data to solve problems.
2. **AI Literacy:** The technical knowledge, durable skills, and future-ready attitudes required to engage with, create with, manage, and design AI systems while critically evaluating their benefits, risks, and ethical implications.
3. **AI Fluency:** The practical ability to effectively leverage AI as a collaborative tool through hands-on experience, combining technical understanding with skilled application in real-world contexts.

School districts and individual schools should feel empowered to modify the timeline below to meet the needs of their students. These are new technologies and each school's readiness for this technology will be different. It is recommended to pilot AI technologies with students before engaging in larger rollouts.

Note: This table presents one possible framework for discussion. Districts should adapt these guidelines based on their specific context, student populations, and local curriculum.

PreK - 2

Students in grades PreK – 2 should not use AI chatbots.

Teachers may use tools that incorporate AI, such as literacy screening tools or tools for developing instructional materials.

Grades 3 - 5: AI Awareness

Students should not use open-ended AI chatbots. AI may be appropriate only when embedded within curriculum-specific educational applications where the AI's responses are constrained to support specific learning objectives and aligned with instructional content.

Teachers may demonstrate or use AI on behalf of students as they teach AI awareness.

Students at this age are still developing their ability to distinguish between technology and human intelligence. They may anthropomorphize AI tools, attributing human-like consciousness or emotions to systems that operate through pattern matching. Additionally, since their foundational literacy skills are still developing, introducing tools that can produce writing far beyond their current ability may bypass the essential struggle needed to develop these core competencies. The focus should remain on building strong reading, writing, and critical thinking foundations that will enable effective AI collaboration later.

Grades 6 - 8: AI Literacy

Students can use age-appropriate, education-specific chatbots under a teacher's direction. Focus on technology that lets the teacher structure the activity and doesn't provide students with access to generic chatbots outside of class.

All interactions should be scaffolded to support the development of AI Literacy.

At this age, students are developing abstract thinking capabilities and can begin to understand more complex technological concepts. They can engage with structured AI tools while building critical thinking skills about technology's role in their lives. However, they still need significant scaffolding to develop appropriate skepticism about AI outputs and to understand ethical implications.

Students are developing their foundational literacy and logical reasoning skills; unstructured AI usage may serve as a crutch, bypassing the struggle necessary for the learning process.

Grades 9+: AI Fluency

Students should begin to leverage AI to support their learning as a cognitive assistant tool. Students should start to use AI for research, learning and co-creation.

In addition to students having a foundation in AI Literacy, schools will need to develop systems to support student AI usage. Schools need to build systems to support academic integrity and teachers need time to adjust assessments to respond to student usage of AI. More info can be found under the section on [Academic Honesty](#).

At this age, Students have developed sufficient metacognitive abilities to reflect on their own thinking and evaluate complex information. They can understand abstract concepts like bias, fairness, and ethical reasoning. They're also developing skills they need to prepare for a workforce where AI usage will be expected. They can handle the cognitive load of managing AI as a thinking partner while maintaining intellectual ownership.

The Four Principles Framework

This guide is organized around four principles that categorize how AI can be utilized in education. This is based on work by SREB's guidance for the use of AI in educational settings (Southern Regional Education Board, 2025).

1. Using AI to streamline the administrative and planning work of education
2. Using AI to support personalized learning
3. Developing students' AI Literacy
4. Using AI-infused tools to develop more cognitively demanding tasks that increase student engagement with creative problem-solving and innovative thinking.

Principle #1: Use AI to streamline administrative and planning work

The first of these principles is using AI to streamline administrative and planning work.

This is not the first because it is the most important or powerful; rather, it is first because it is the best place for teachers and staff to first gain experience using AI.

Many of these tasks avoid using student data, and they provide ample time for the educators to review anything generated by AI, making these the ideal use cases for utilizing AI for the first time. Efficient use of time on administrative tasks can enable educators to spend more time working directly with students and collaborating with colleagues.

Opportunities Without Student Data

AI as a Planning Partner

Generative AI tools allow teachers to have a planning partner with vast background knowledge, excellent writing skills, and the ability to read and analyze lengthy materials. Some of these use cases include:

- Assistance with aligning class activities to curricular standards
- Editorial review of draft writing
- Brainstorming activities and assignments

Improve and Enhance Current Assignments

Educators can use AI tools to personalize assignments and ask for ideas to increase cognitive demand, differentiate them for specific learning needs, or provide more interactive activities. With the click of a few buttons, teachers may also use these tools to develop slides and graphic organizers or to develop questions based on a reading, image, or video. AI tools can also create samples of student work based on the assignment that students can utilize for practicing fact-checking, grammar and tone editing, and giving feedback.

Support with Other Aspects of Teaching

While lesson and unit planning are essential aspects of a teacher's job, teachers also have a myriad of other tasks they need to accomplish each day. AI can act as an editor, assisting with the tone of emails, editing parent newsletters, and proofreading reports. It can also support multilingual English learner students by translating assignments and other learning materials.

Differentiate Course Materials

Educators can develop worksheets and readings at different levels or focus on the different interests of their students. AI is a tool that can help educators better differentiate instruction to meet each individual student's personal learning needs.

Opportunities With Student Data

Caution: Never put student data into an AI system unless the district's IT department has vetted that system and has authorized its use in this way.

While most systems today don't interact directly with district data systems, this will change in the next few years, with SIS (Student Information System), LMS (Learning Management System) and MTSS (Multi-Tiered System of Supports) platforms building AI integrations directly into their systems enabling new tools for teachers and administrators that leverage rich and up to date information about students.

Administrative Tasks Related to Student Support

By mining student data, AI can make many processes more efficient. For instance, a teacher who is preparing personalized updates for parents and guardians can have AI draft individualized updates, making use of the available data. Administrators who want to understand their data can use natural language to have a conversation about the meaning of their data.

Care must be taken when utilizing AI to analyze data. AI systems can make mistakes; the analysis provided by the AI system should always be verified against the educator's own understanding of the data.

Providing Feedback on Student Work

AI can instantly provide students with detailed feedback on early drafts of a writing assignment. Teachers can set up the AI system by providing assignment instructions, rubrics, examples of student work, and appropriate feedback to provide the AI system with the context necessary to accomplish the task. The AI can provide feedback on writing style/quality, supply students with reflective questions to consider for future drafts and encourage deeper analysis and inquiry.

The main advantage of using an AI is the speed at which students can receive feedback to improve the writing process. This can be incorporated into the writing process and combined with writing conferences to support students over multiple drafts.

This strategy helps model effective AI use to support learning. It also helps discourage students' use of AI without permission, which often interferes with the learning process.

Notetaking

AI tools can be utilized to take class notes as well as notes in meetings. Class notes can help meet accommodation needs and support students who missed class. Meeting notes can make meetings more effective and allow for clear communication and follow-up.

Teacher Development

Video recordings of class can be paired with AI analysis for teachers to be able to reflect on their teaching and work with instructional coaching.

Data Analysis

Generative AI allows for English language conversations about student data. This can empower educators and administrators who may have a difficult time navigating and analyzing the large amounts of student data in their systems. Because the potential for inaccurate analysis exists, it is important to use systems that are purposely built and to verify any AI analysis.

Cautions

AI Bias

Generative AI systems are inherently biased, based on their training data. The output generated by AI should be carefully reviewed. [Learn more about mitigating AI Bias.](#)

Student Data Privacy

Federal and state laws mandate that districts and schools protect students' data and privacy. [Learn more about safeguarding student data privacy.](#)

Student Intellectual Property

AI systems, not vetted by school districts, may be training their AI model using data from chat conversations. [Learn more about safeguarding student data privacy.](#)

Cognitive Offloading

When utilizing AI for administrative tasks, educators should ensure that automation enhances rather than replaces their core teaching practices and professional judgment. [Learn more about cognitive offloading.](#)

Fact-Checking and Reviewing AI Output

Educators must carefully review all AI-generated output. AI systems demonstrate strong writing capabilities, which may obscure subtle errors in logic. Generative AI models do remember precise information, which can result in the model making up facts, including sources. All facts and references generated by AI require verification. Providing

reference documents to the AI model and enabling internet access can help minimize such hallucinations.

- Cross-reference with trusted sources: When AI provides statistics, historical facts, or scientific information, verify against curriculum materials, textbooks, or reputable websites.
- Be especially cautious with: Dates, names, specific statistics, quotes, and citations. These are areas where AI frequently makes errors.

When to Avoid Using AI

There are important cases when the use of AI is inappropriate. Always be transparent with AI usage and have active discussions amongst educators about when staff should avoid using AI. Some of the situations that need special review or should be avoided entirely include:

- High-stakes grading or assessment decisions - Summative grades should fully reflect educators' professional judgment and not be influenced by the biases of AI.
- Sensitive student situations - Communications about discipline, special needs, or family circumstances require human sensitivity.
- Professional evaluations - Writing observations of colleagues or staff recommendations.
- Legal or compliance documentation – Districts should seek legal advice before approving AI usage for IEP goals, 504 plans, or mandated reporting situations¹.
- More discussion about appropriate AI usage for IEPs and 504 plans follows in the next Principle.
- Medical-related information – Unless the IT Department has specifically vetted an AI solution for HIPAA compliance, staff should avoid entering any medical-related information into an AI system.

Principle #2: Use AI to support personalized learning

AI tools, built specifically for education, can support personalized instruction by adjusting to the specific needs of each student. AI creates new opportunities for educational technologies in this space, and new products will be developed that leverage AI to support students based on their individual needs. AI can serve as a teaching assistant that can individually work with learners.

¹ [CDT - The Use of Generative AI to Develop Individualized Education Programs for Students with Disabilities](#)

Opportunities

Tutoring Systems

AI tutoring systems can provide one-on-one tutoring and coaching for students. Intensive human tutoring has shown large and consistent benefits, but it is hard to scale these services (Barshay, 2025). There is huge potential for AI's usage in these scenarios, and some early pilots have been very successful.

AI tutoring needs to be carefully developed, with the AI following effective tutoring practices. When schools are evaluating AI solutions, they should look for systems that:

- Have access to course and curricular materials, ensuring that jargon, explanations, and scaffolding match what is used by the teacher in class
- Are designed with tutoring best practices
- Have research data backing up the effectiveness of the solution
- Fine-tuned models that are trained on exemplary tutoring interactions are likely to outperform generic frontier models
- Further information on vetting AI solutions is available in the [Tool Vetting Framework](#) section of this document.
- Additional resources on AI tutoring include:
 - Indiana's report on a large-scale AI tutoring pilot involving multiple vendors²
 - Khan Academy's insights from the development of their AI tutor³
 - Insights from Stanford's use of AI tutoring⁴
 - Insights from a Harvard study of AI tutoring⁵
 - A meta-analysis of K12 AI tutoring systems showing mixed results⁶

Supporting Students with Special Education and Accommodation Needs

² [AI-Powered Platform Pilot Grant Final Report Indiana Dept of Education](#)

³ [Khanmigo Math Computation and Tutoring Updates - Khan Academy Blog](#)

⁴ [How AI can improve tutor effectiveness](#)

⁵ [AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting](#)

⁶ [A systematic review of AI-driven intelligent tutoring systems \(ITS\) in K-12 education](#)

AI tools can significantly improve personalized learning by tailoring educational content to meet the unique needs and preferences of each student, especially those with disabilities (Perez, 2024). COSN and CAST developed an excellent report covering these opportunities, including coverage of risk and limits using these tools⁷.

Some examples include:

- **Note-taking and live captioning:** AI can be utilized for live captioning and for notetaking purposes. When utilizing these technologies, work with district technology staff to identify solutions that meet the privacy needs of schools, as many of these solutions are not built specifically for education.
- **Visual recognition systems:** OCR technology has existed for a long time, but modern multi-modal AI can make sense of the images, graphics, and layout on the page as well. Additionally, the AI systems that read the text out loud have natural intonation and inflection.
- **Translation and leveling of readings:** AI technology can level readings, producing versions with different reading levels. It can also be utilized to translate texts, with superior results compared to older systems. When translating texts, mistakes can occur. One technique for verifying translations is to retranslate the text into the original language to see if any meaning was lost in translation, make sure you utilize a new chat session when employing this technique.

As seen above, there are many useful ways AI can be used to implement universal design of learning (UDL), specially designed instruction (SDI), and accessibility features for equitable access to material. In addition to these uses, AI can be used to help with special education documentation (e.g. IEPs, 504 plans); however, using AI in this way presents significant concerns that should be addressed before implementation. IEPs and 504 plans are the foundations on which special education services stem from, and using AI poses a danger of losing essential professional assessment and breaching of personally identifiable information (PII) protection. Using AI to generate IEPs or 504 plans also risks taking away from the *individualized to the student* requirement of this special education documentation, especially considering the issues with bias in AI training data. It is important to remember that IEPs are legally binding documents that describe the educational services and rights a student will receive, as well as being legally required to be a collaborative development process including the student's education team, parents/guardians, and specialists.

Keeping those cautions in mind, using AI tools to help with IEP and 504 plans, as well as implementation of special education services, is possible and can be very effective. The most important step in using AI tools in this context is to have human oversight every step of the way. For example, IEP goal generation can be done using AI, but the

⁷ [AI & Accessibility in Education](#)

prompt and content to generate the goal requires a secure FERPA compliant system and should be reviewed by the IEP team for relevance to the student's capabilities and needs. As with all uses of AI, but especially with students in this context who need equitable access to education, technology should enhance, not replace, the implementation of special education services.

Predictive Analytics

AI systems can provide predictive analytics in K-12 education by analyzing large amounts of student data to identify patterns, forecast future outcomes, and enable proactive interventions. This helps educators, administrators, and even students themselves make more informed decisions. As an example, a predictive analytics system that has access to high school attendance and grade data could automatically notify a student's advisor that there is a concern.

Automated Feedback

"Feedback is the most powerful single moderator that enhances achievement" - John Hattie.

The quality and speed of feedback that students receive have a large impact on their learning. AI provides a tool to automate this feedback. This could be instantaneous (no teacher review) or could be used by the teacher to shorten the time it takes them to generate feedback on student work. Often, AI systems can provide more in-depth feedback on writing than teachers, because teachers lack the necessary time to provide the same level of detail. This opens new opportunities to engage in the development process, helping students through multiple drafts and revisions.

Adaptive Learning Platforms

Curriculum developers are building AI into their platforms. This allows for an adaptive system that adjusts the materials presented to the student based on the student's performance on activities and personal interests.

Language Translation

Large language model AI systems excel at language translation, preserving the meaning and the style of the original text. Coupling this with AI powered voice generation and audio transcription allows for natural conversations to be translated live in real time. This opens new possibilities for supporting students and caretakers whose native language is not English. Special attention is needed to review AI outputs, especially if the staff member is not an expert speaker of the language. One technique for evaluating outputs is to use a fresh chat conversation (new context window) to translate the AI output back into the original language.

Principle #3: Develop students' AI Literacy

AI Literacy refers to the ability to understand, evaluate, and use AI. Several frameworks exist for teaching AI literacy, including Digital Promise's AI Literacy Framework⁸, the AILit Framework⁹, and CSTA/AI4K12's AI Learning Priorities¹⁰. This guidance draws on Digital Promise's framework for its accessibility and alignment with Vermont's existing ISTE Standards adoption. Districts interested in deeper computer science integration may find value in exploring AI4K12's approach.

What is presented here should be viewed as an example approach and is not meant to be prescriptive.

Understanding AI

Understanding AI means knowing how it works, not through metaphors like "thinking" or "learning"; rather, it should be understood mechanistically. Generative AI systems use data, apply statistical models to identify patterns, and generate outputs by predicting what comes next. Students benefit from understanding where training data comes from, how models are built, and why outputs can sound confident despite being incorrect.

Elementary School Example

Using Google's Teachable Machine, students train a simple image classifier to recognize different hand gestures. Students quickly discover that training requires far more examples than they expected: a few photos aren't enough. The model needs dozens of images from different angles, distances, and lighting conditions to work reliably. When it fails, students can diagnose the problem and improve the training data themselves. This hands-on experience builds intuition for how AI learns from patterns in data rather than from understanding, and why robust training requires more data than most people assume.

Secondary School Example

In a social studies unit, students examine how algorithmic curation shapes public discourse. The class studies documented examples showing how trending algorithms can surface different stories to different communities. Students analyze primary sources: excerpts from Congressional testimony by tech executives, summaries of academic research on filter bubbles, or journalist investigations of algorithmic amplification. Students discuss the civic implications that arise when citizens in the same community encounter different information landscapes.

⁸ Digital Promise's [AI Literacy: A Framework to Understand, Evaluate, and Use Emerging Technology](#)

⁹ [AILit Framework – Empowering Learners for the Age of AI](#)

¹⁰ [AI Learning Priorities for all K-12 Students](#)

Evaluating AI

Evaluating AI means critically assessing outputs for accuracy, bias, and appropriateness. This includes asking: Is this information correct? Whose perspectives are represented or missing? Is AI the right tool for this task?

Students' skills evaluating AI develop most naturally through disciplinary practice. Nick Potkalitsky describes this as the "knowledge asymmetry problem"¹¹: students can only catch AI errors in domains where they have genuine expertise. When students compare an AI-generated historical summary with primary sources, they strengthen both historical thinking and AI evaluation skills simultaneously. When they check an AI explanation of a scientific concept against their experimental observations, they deepen both their science understanding and their critical stance toward AI.

Elementary School Example

A third-grade science class observing plants might record what they notice (leaf shape, stem characteristics, flower placement), then compare their observations with an AI-generated plant description. What did AI get right? What did it miss? What can only be known through direct observation? This teaches students that AI predicts based on patterns while scientists verify with evidence. This is a foundational AI literacy concept embedded naturally within existing science standards.

Secondary School Example

In a social studies class studying the American West, the teacher might ask AI to describe life in frontier towns or generate images of the American West, then have students examine the response against primary sources like diaries, photographs, and census records. Students learn that AI reflects patterns in its training data, developing biases and stereotypes, which can overrepresent popular narratives and underrepresent marginalized perspectives.

Using AI

Using AI means making informed decisions about when and how to employ AI tools. Digital Promise identifies three types of use: interacting with AI, creating with AI, and problem-solving with AI.

The skills of using AI line up with the ISTE Student Standards, which have been updated to cover AI usage for learning. Many Vermont schools will be able to integrate the skills of using AI into their existing digital literacy curriculum.

Secondary School Example

A teacher might provide access to an AI writing assistant configured with the

¹¹ [The Knowledge Asymmetry Problem: Why Student Expertise Matters in AI Collaboration](#)

assignment rubric and class expectations. After completing the first draft independently, students submit their draft for AI feedback, then reflect: Which suggestions improved clarity without changing my ideas? Where did the AI miss my intent? How do I keep my authentic voice while incorporating useful advice? This builds metacognitive awareness about use of AI tools.

Connection to Vermont's Educational Priorities

This approach to AI literacy directly supports Vermont's focus on improving outcomes in literacy, mathematics, and science. The relationship is reciprocal: content knowledge enables meaningful AI evaluation, while practicing AI literacy deepens engagement with content.

At the same time, AI literacy prepares students for a world where these tools are ubiquitous. Students who understand how AI works, can evaluate its outputs critically, and make informed decisions about its use are better positioned to thrive in higher education, careers, and civic life.

Principle #4: Use AI-infused tools for cognitively demanding tasks

These use cases are generally not recommended for elementary students, and in many cases, middle school students. See [developmental appropriateness of AI](#) for details.

Overview

AI opens opportunities for students to engage in sophisticated intellectual work that would be difficult or impossible to accomplish independently. This principle goes beyond using AI as a tool; it positions AI as a thinking partner that enables students to tackle complex problems, engage in deeper analysis, and explore ideas at a level of sophistication previously reserved for advanced study.

When implemented effectively, AI becomes a cognitive amplifier. Students can engage in Socratic dialogues where AI challenges their reasoning, participate in complex simulations and debates with an adaptive opponent, teach concepts to an AI that asks probing questions, and reflect on their learning with a partner that helps identify patterns and blind spots. These interactions require students to think more deeply, defend their reasoning more rigorously, and examine their assumptions in ways that traditionally required one-on-one mentorship from a teacher or subject expert.

This approach fundamentally shifts what we can expect from student work. When students have access to an AI thinking partner, the bar for acceptable work rises significantly. A research paper should demonstrate more sophisticated analysis. A problem-solving exercise should tackle greater complexity. A creative project should show a deeper exploration of possibilities.

However, this principle requires careful implementation. Students need strong foundational knowledge and well-developed AI literacy skills before they can effectively collaborate with AI on complex tasks. The goal is to develop students who can leverage

AI to extend their intellectual reach while maintaining ownership of their learning and demonstrating genuine understanding. This is most appropriate for grades 9-12, where students have developed the metacognitive skills necessary to engage AI as a true intellectual partner rather than a crutch.

In addition, teachers should incorporate formative assessment structures for assessing student learning. In class conferencing, student presentations, class discussion and student reflections should be used to focus on what the student learned from the activity.

Opportunities

Interactive Learning Partnerships

AI can engage students as a Socratic partner, challenging their thinking through targeted questioning and counterarguments. Students defend complex positions, identify flaws in reasoning, and navigate philosophical or ethical dilemmas with AI pushing their logic. This goes beyond simple Q&A—students must articulate nuanced arguments and respond to sophisticated challenges that adapt to their reasoning.

Teaching Through Role Reversal

Students solidify their understanding by teaching concepts to an AI that plays the role of an intelligent but naive learner. The AI asks probing "why" and "what if" questions that reveal gaps in student understanding, forcing students to examine assumptions and edge cases. For example, a student explaining a physics concept must respond to AI questions about real-world applications and theoretical limits, deepening their own comprehension through the act of teaching. It is helpful to use a well-crafted AI prompt like the one Ethan Mollick has shared¹².

Complex Simulations and Debates

AI can serve as an adaptive opponent or partner in sophisticated scenarios. Students might argue historical positions against an AI representing opposing viewpoints with historically accurate counterarguments, negotiate complex business deals where AI adapts its strategy based on student moves, or navigate evolving ethical dilemmas where AI introduces complications that challenge initial moral stances. These interactions develop critical thinking and argumentation skills at levels difficult to achieve through traditional assignments.

Collaborative Problem-Solving and Creation

Students work with AI to tackle problems beyond their independent capabilities. In scientific inquiry, AI helps generate hypotheses and analyze data while students design experiments and interpret results. For complex writing projects, students produce initial

¹² [Teach the AI as student prompt from Ethan Mollick](#)

drafts independently, then collaborate with AI to identify argumentative weaknesses and explore alternative perspectives. The key is that students maintain intellectual ownership while using AI to reach higher levels of sophistication.

Metacognitive Reflection Partnerships

Beyond basic reflection, students engage AI in a conversation, where the AI's questions lead to students analyzing their own learning processes. AI helps identify patterns in problem-solving approaches, cognitive biases in reasoning, and gaps in understanding. Students develop learning plans based on these insights, building self-awareness about their thinking processes. This develops the metacognitive skills essential for lifelong learning. Ethan Mollick shared a prompt that can be used by educators for this activity¹³.

Project-Based Learning Enhancement

AI amplifies what students can accomplish in extended projects. Engineering students use AI to generate design alternatives and run simulations while making critical design decisions. History students process large sets of primary sources with AI assistance but construct their own historical arguments. The AI handles computational or analytical tasks that would be prohibitively time-consuming, allowing students to focus on higher-order thinking and synthesis.

Concerns

Cognitive Bypassing and Over-reliance

Students may become dependent on AI for intellectual work, missing critical learning that comes from productive struggle. [Learn more about cognitive offloading](#).

Assessment and Authenticity Challenges

Impressive student work may mask shallow understanding, as students might use sophisticated language they don't fully comprehend or accept AI suggestions without critical evaluation. [Learn how to address academic honesty concerns](#).

Readiness and Equity Gaps

Students with stronger foundations may leverage AI more effectively. Schools should carefully consider [grade-level appropriateness](#), provide explicit instruction in AI collaboration techniques, and potentially pilot approaches with advanced students before broader implementation.

¹³ [Metacognitive reflection prompt from Ethan Mollick](#)

Part 3: Responsible AI Usage

Risk Mitigation Strategies

Risk of Doing Nothing

Schools that avoid addressing AI usage expose themselves to a different set of risks. Students are more likely to use AI systems outside of school without any guidance or AI literacy training. Staff may utilize public AI systems in ways that risk data privacy. Students will graduate unprepared for a world that is being radically transformed by AI. Doing nothing also introduces equity and access issues, as some students have access to strong AI models and develop the skills to use them, while others do not.

The risks of using AI described below are real; however, this is not a reason for inaction. There are steps that can be taken to minimize each of the concerns below.

Academic Honesty

Academic honesty concerns are the single most cited concern related to AI amongst school leaders. Students, who may have previously reached out to friends or siblings, can now turn to AI for help. Despite this, an ongoing research study by Stanford researchers suggested that cheating amongst U.S. High Schoolers did not increase with the release of ChatGPT (Spector, 2023).

Schools need to have a consistent strategy for promoting academic honesty; it is much more difficult for each teacher to address these issues independently. Developing these strategies are important discussions for school leaders to facilitate amongst their faculty.

AI can be utilized on assignments in multiple ways. It is not as clear-cut as simply saying, “AI usage is not allowed.” AI is being baked into the word-processing applications that students utilize every day. Does using a grammar check count as AI usage? What about Grammarly, a product that goes beyond the classic grammar check? What about a student who uses AI to help conduct research or to explain a concept to them?

To respond to these concerns:

- Be clear that there are times when the use of AI by students is inappropriate. Teachers should feel empowered to tell their students that the use of AI is inappropriate in these situations.
- Review academic honesty policies with respect to AI. AI for education has a helpful short guide for drafting an AI academic policy at your school¹⁴.

¹⁴ [Drafting a GenAI Academic Policy at your School — AI for Education](#)

- Discuss how students should cite AI usage. Both MLA and APA now have formats for citing AI use. Educators should inform students whether they should save and submit their AI chat logs.
- Foster transparency by having educators lead by example, modeling transparency and academic honesty about their own use of generative AI tools.
- Encourage teachers to focus on "process" over "product." Students often seek help in getting the product completed, but this is the mere artifact of the learning process. Teachers can focus on the process by asking students to reflect on their learning, providing in-class one-on-one feedback on early drafts or stages of work, and asking students to show and explain their work. Teachers should include this "process" as part of their assessment and grading, which refocuses students on the learning process and not simply completing the "work."
- Determine how teachers currently handle AI usage and discuss a common language that can be utilized. Instead of accusing a student of cheating, these situations are best handled with curiosity, asking more about the student's process and how they developed their product.
- Encourage teachers to move more assessments into class time and not use homework as the sole means of grading. Beyond tests and quizzes, this can include formative assessments like writing conferences, class presentations, structured class discussions, reflective writing prompts, etc. Activities like this support active learning and are good teaching practices.
- Consider AI-resistant assignments. Assignments involving reflection or discussion of class experiences cannot easily be completed by AI without effort from the student.
- Treating the academic honesty issue as a cry for help from students can be very effective. Most often, students in these situations are confused by the material and overwhelmed by the pressure to stay on top of their coursework. They need to be provided with a means to catch up with the class and support to better understand the course content.
- Do not rely on AI auto-detection tools. These tools are not reliable enough to be used in academic honesty disputes, sometimes show false positives, and they are easily bypassed by students.
- Developing consistent messaging to explain how AI usage can occur on an assignment is important. Many schools have discovered that a more nuanced scale of AI usage is preferable to a simple traffic light system. One of the better-known examples is the AI Assessment Scale (AIAS). Lamoille South Supervisory Union developed their own Student Generative AI Acceptable Usage Menu for their teachers, which is presented here as a positive example that provides clear guidance for teachers. Schools are encouraged to develop their own scales, using either the AI Assessment Scale or LSSU's menu as a starting point.

DRAFT - Student - Generative AI Acceptable Usage Menu				
				
No Use	AI as a Thought Partner	AI as an Editor	AI as an Analyst	AI as a Co-Creator
AI is not permitted on this assignment for any reason	AI can be used only for initial inquiry, brainstorming, and to help generate ideas.	AI can be used to edit and receive feedback on original content.	AI can be used to analyze data, identify patterns, and identify themes.	AI can be used to help complete assignments and projects.
Reading: You must complete all reading without any help from AI. Content Creation: You must complete the assignment using only your own knowledge, skills, and Creativity. Problem-Solving: You must solve all parts of assigned problems using only your own knowledge, skills, and reasoning.	Reading: You must read the original text as assigned, but AI may be used to help you understand the text. Content Creation: You can use AI tools only for initial idea-gathering, and not to complete the assignment. Problem-Solving: You must generate your own solutions to problems, but may use AI to suggest strategies, explore concepts, and provide helpful ideas.	Content Creation: You can use AI feedback to help edit and revise your original content. Problem-Solving: You may use AI to improve the clarity, organization, and simplicity of a solution you have created, but not to correct errors.	Reading: AI may be used to help you analyze the reading, but you must read the text assigned. Content Creation: AI can be used to help you create visualizations or graphs to help identify new patterns or themes, but you must complete the assignment Yourself. Problem-Solving: You may use AI to help identify mathematical processing errors and misapplication of principles, but you must correct your work by yourself.	Reading: You can create and read an AI summary instead of reading the original text. Content Creation: You can use AI to create parts of an assignment as permitted. You need to review AI-created work to ensure its accuracy. Problem-Solving: You may use AI to generate solutions when solving problems. You must check and verify your solution.
No disclosure statement required	Students must disclose AI usage. Copies of or links to chat(s) must be submitted	Students must disclose AI usage. Copies of or links to chat(s) must be submitted	Students must cite AI created content in the appropriate content. Copies of or links to chat(s) must be submitted	Students must cite AI created content in the appropriate content. Copies of or links to chat(s) must be submitted

LSSU; last revised 6/18/25
Based on the Kinkaid School GenAI Acceptable Usage Menu

[Student Generative AI Acceptable Usage Menu](#) from Lamoille South Supervisory Union – from 6/18/25 version

Cognitive Offloading

AI gives us the ability to "cognitively offload" tasks, essentially outsourcing our thinking to a tool. For professionals, this is a powerful way to increase efficiency and yield a superior final product. For students, however, offloading the wrong tasks can interfere with learning and skill development.

The central challenge for educators is to determine when AI serves as a valuable learning tool and when it becomes a crutch that hinders skill development. As author and researcher Ethan Mollick discusses in his post, "Against Brain Damage,"¹⁵ improper use of AI can inadvertently prevent students from engaging in the productive struggle necessary for learning. Just as we wouldn't want a student to use a calculator when they are first learning multiplication and division, there are situations where AI usage, particularly open unstructured AI usage, is harmful to the learning process. Teachers should feel empowered to say when AI usage is inappropriate.

¹⁵ [Against "Brain Damage" - by Ethan Mollick](#)

The decision to use AI should be driven by the lesson's learning objectives. Before assigning a task involving AI, teachers should ask:

1. What are the primary skills I want students to develop with this activity?
2. Will AI's role in this task help them practice that skill, or will it perform the skill for them?

This leads toward a structured approach when selecting which AI to use:

- Education-specific AI platforms tailored to your activity or AI, with an initial prompt created by the teacher, can provide a structure to ensure that AI is supporting the learning process
- When students use general-purpose AI, teachers need to be able to structure and teach students how to use these AI systems. This requires a greater level of AI literacy for both the teacher and the students.
- We need to teach students to lean into the hard work of learning. We know from research that learners are not always the best judges of how well they are learning a topic. Vox created a great video that explores this topic through a series of interviews with students and teachers¹⁶.

Data Privacy

Schools need to protect student data, conforming to both FERPA and COPPA. In addition, generative AI uses data to train AI systems, and schools should be concerned about the intellectual property of students being utilized to train these systems.

Providing teachers and students with approved, secure AI systems is one of the best ways to protect student data. A separate section of this guide covers [tool vetting](#) in depth. When schools do not provide official tools and clear instructions for these tools, teachers and students will be tempted to utilize public AI systems instead.

It is not enough to remove student-identifiable information before submitting student work to an AI system. This still exposes the student's intellectual property to the system. SU/SDs should have a privacy agreement with vendors that include promises not to train AI models based on chat conversations. This is the case for education-specific AI systems as well as generic systems with AI versions (Gemini for Education).

Health Insurance Portability and Accountability Act (HIPAA) data (medical data) is a particular concern, as it has requirements that are not a part of FERPA or COPPA. Most educational AI systems have not been designed to comply with HIPAA. In many cases, it is simpler for schools to avoid the use of AI for HIPAA protected data. In the future, it

¹⁶ [AI can do your homework. Now what? – Vox Video](#)

is likely that there will be use cases that involve the use of AI, but initially, it may be simpler for schools to avoid the use of AI with HIPAA protected data.

Mental Health Risks

There are real concerns that people can form unhealthy relationships with AI systems. Kids, who may be seeking companionship, can find these systems very appealing. As these systems integrate voice communication and avatars, this may be a part of a growing problem. Common Sense Media¹⁷ reports that 72% of U.S. teens have used AI companions at least once (Common Sense Media, 2025) and they recommend that teens avoid using AI companions¹⁸.

The same report outlines concerns for these companion style apps, with one third of teens using AI for social interaction and relationships. For example, 12% report that they can say things to AI that they can't talk about with friends and family. The report asks that educators take the lead on digital literacy to help kids safely navigate an AI-powered world. The APA (American Psychological Association) issued a health advisory¹⁹ on similar notes, which outlines these concerns.

AI is not a substitute for professional mental healthcare²⁰. While nearly half of students with mental health challenges report using AI for counseling²¹, these systems lack the nuanced empathy, cultural competence, and crisis-response capabilities of trained professionals. School staff should emphasize that 'simulated empathy' cannot replace the genuine human connection and clinical expertise found in school-based services. To ensure student safety, staff must proactively direct students to school counselors, social workers, or psychologists for confidential, evidence-based support.

School leaders may need to navigate students having unhealthy interactions with AI outside of school time and need to support families navigating these issues. One example is the increasingly popular toys for young kids with AI embedded in the product. Kids can form intense relationships with these toys when they do not yet have

¹⁷ [Talk, Trust, and Trade-Offs: How and Why Teens Use AI Companions | Common Sense Media](#)

¹⁸ [Common Sense Media ratings for Social AI Companions | Common Sense Media](#)

¹⁹ [Artificial Intelligence and Adolescent Well-being: An APA Health Advisory](#)

²⁰ [Common Sense Media ratings for AI Chatbots for Mental Health Support](#)

²¹ [AI Research — Sentio University](#)

the maturity to recognize the difference between a machine and a living thing²². In many cases, this will happen on systems not internally vetted by the school system.

As a preventative, it is important to include these topics in the AI literacy training provided to students as well as caretakers. The [parental & guardians guidance](#) section of this document has some additional resources. Schools can best safeguard students by combining these AI literacy efforts with a positive school climate²³, a focus on belonging and healthy relationships, and a robust multi-tiered system of support (MTSS)²⁴.

Student Well-Being and Safety

AI can be used for misinformation, and these uses can threaten students' well-being and safety. Deepfakes and non-consensual intimate images (NCII) can be generated to malign and cyberbully students.

The Center for Democracy and Technology issued a report, "In Deep Trouble: Surfacing Tech-Powered Sexual Harassment in K-12 Schools"²⁵. The report identified non-consensual intimate images as a significant problem in K12. Female and Lesbian, Gay, Bisexual, Transgender, and Queer/Questioning (LGTBQ+) students were the most concerned about non-consensual images and had lower confidence in their school's ability to prevent and respond to deepfakes. (Report – In Deep Trouble: Surfacing Tech-Powered Sexual Harassment in K-12 Schools, 2024)

School leaders should assess what policies in their student handbook and district policies cover this behavior, and how they can potentially respond. The Center for Democracy and Technology released a model policy and resources²⁶ that can serve as a resource.

IT & school leaders should consider what tools are at their disposal to respond. Instagram has a school partnership program²⁷ to provide school leaders with priority

²² [‘I love you too!’ My family’s creepy, unsettling week with an AI toy | Artificial intelligence \(AI\) | The Guardian](#)

²³ [School Climate - Agency of Education](#)

²⁴ [Vermont Multi-Tiered System of Supports](#)

²⁵ [In Deep Trouble: Surfacing Tech-Powered Sexual Harassment in K-12 Schools - Center for Democracy and Technology](#)

²⁶ [Model Policy and Infographic: Non-Consensual Intimate Imagery \(NCII\) for K-12 Schools - Center for Democracy and Technology](#)

²⁷ [Partnering with Schools to Create a Safer Online Community - Instagram](#)

tools for reporting cyberbullying. While this tool could be construed as endorsing Instagram, it should be considered as school leaders often need to respond to incidents that happen outside of the school's control.

Students will need to navigate a world increasingly populated by AI-generated content. It is imperative that schools teach digital literacy skills to navigate this world and provide students with tools to separate fact from fiction. The ISTE Student Standards²⁸, which Vermont adopted in 2017, require students develop the skills to evaluate information (1.3.b) and safeguard their well-being (1.2.c). AI for Education developed a classroom guide and discussion questions²⁹ that could serve as a starting point for teachers designing lesson plans.

Bias and Fairness

Generative AI models are trained on huge corpuses of data; in the case of large language models, this includes all quality text on the internet as well as other text the AI company can access. Baked into this text are a lot of pre-existing biases and stereotypes. The text is also detached from real world experience. AI companies further train their AI on sample conversations, and while these are meant to mitigate bias, they can often introduce unforeseen outcomes. These factors lead to models that incorporate biases and stereotypes. In addition, generative AI is probabilistic in nature, always generating different output even given the same input.

Educators and students need to use these models, keeping these potential biases in mind. AI systems are excellent writers, which means that they sound convincing even when they are incorrect.

Some mitigation techniques include:

- Provide a way for teachers and students to provide feedback on the AI system. Survey teachers and students to determine if they encounter bias when using an AI system.
- Provide AI literacy training for staff & students, including digital literacy training covering how to verify data.
- Ensure that a human is responsible for reviewing the output (a separate section addresses human oversight)

To explore bias, stereotyping, and logical blind spots in AI systems, consider the following prompts:

²⁸ [ISTE Student Standards](#)

²⁹ [Uncovering Deepfakes: Classroom Guide — AI for Education](#)

- Ask the AI to generate images of people in a role or job. What stereotypes were built into the generated images?
- Ask the AI a question that needs real-world knowledge that is unlikely to be found in written text. Example, “I hang 4 towels on a laundry line, and it takes four hours for the towels to dry. How long would it take for 10 towels to dry on the laundry line?” AI systems struggle to answer problems like this (they see the pattern of a typical math problem and jump to an incorrect conclusion).
- Ask the AI a question that relies on how the data is presented to the AI. For instance, when a document is provided to AI, it is translated into tokens that the AI understands and information can be lost in this translation. Older and poorly designed AI systems struggle with “How many times does the letter ‘a’ appear in banana,” and struggle with the images and charts that appear in documents.

Equity and Access

AI opens new opportunities to address equity and accessibility (see [Supporting Students with Special Needs and Accommodations](#)), but the introduction of AI in education does create some unique equity and access issues that schools should address.

Home Access Disparities: Some students have access to paid AI models at home, while others lack reliable internet access entirely. When homework or independent practice involves AI tools, these disparities can create inequitable learning conditions. AI is changing the nature of homework, and equity should be a consideration, not just the concerns of academic honesty violations (see [Academic honesty](#)). When schools implement AI tutoring or AI feedback solutions, home access should be considered from an equity lens.

Parent and Caregiver Experience: Parents with greater experience using AI may better support their children's AI learning at home, while others may be unfamiliar with these technologies. Schools can mitigate this by providing resources and learning opportunities for families alongside student AI literacy initiatives. (see [Parental & Guardian's Guidance](#))

Foundational Skills and Achievement Gaps: Students with stronger foundational literacy, writing, and critical thinking skills will extract more value from AI interactions. Without intentional intervention, AI could widen existing achievement gaps rather than close them. AI literacy training, digital literacy instruction, and structured practice with AI tools can help ensure all students benefit equitably (see [Principle #3 Develop Student's AI Literacy](#)).

Accessibility of AI Tools: AI solutions need to be vetted for accessibility, consistent with any other educational technology product. Because these solutions are relatively new, some vendors have not fully incorporated accessible design. (see [Tool Vetting Framework](#)).

Teacher Readiness: Teacher experience with AI varies widely, with some educators resistant to adopting these technologies. Students in classrooms with AI-proficient teachers may have advantages over peers whose teachers are less comfortable with these tools. Professional development and ongoing support can help ensure more consistent AI integration across classrooms (see [Professional Development Resources](#)).

Ethics of AI Use

There are legitimate concerns about how AI systems are trained and used, which include:

- AI is built on the intellectual property of artists and writers without their consent, compensation and without credit. Major AI developers ran roughshod over these intellectual property rights when they designed these systems. Examples include Meta training their AI model on pirated books³⁰, artwork training AI models without permission³¹ and the lack of sufficient compensation to writers for their work³².
- AI systems are computationally expensive, meaning that a large amount of energy needs to be consumed both in their training and use. The United Nations has developed a useful overview³³ of this issue.
- AI systems have the potential to enable efficiencies that threaten jobs on a massive scale. By 2030, an estimated 92 million jobs will be displaced by AI, according to the World Economic Forum³⁴. In the short term, only select industries have been affected, and mostly for entry-level positions. Disproportionally, this approach affects recent graduates, who need to develop work-based experience. When entry level job productivity increases, companies are inclined to hire fewer of these positions. Schools will need to respond to changes in the job market so that students are prepared for in-demand jobs when they graduate.
- AI systems can empower nefarious actors and oppressive regimes with powerful tools of misinformation and control.

³⁰[The unbelievable scale of AI's pirated books problem](#)

³¹[Viral Lists Reveal Artists Whose Work May Have Trained an A.I. Art Generator](#)

³²[Anthropic destroyed millions of print books to build its AI models](#)

³³[Artificial intelligence: How much energy does AI use? - United Nations](#)

³⁴[World Economic Forum – 2025 Future of Jobs Report](#)

- AI technology may further concentrate power among a few and exacerbate inequity.
- Anthropomorphizing AI risks cheapening the value of each human life.
- There are risks of AI systems getting out of control, pursuing goals that are out of alignment with the best interests of humanity.

As Educators, we can create space for students to explore and discuss these topics. Everyone should be concerned about the potential harm of AI systems. Vermont teachers and students have very strong and legitimate concerns about AI. Educating students about the risks of AI is our responsibility as educators and should be a component of an AI literacy curriculum.

While these are legitimate concerns, the widespread adoption of AI is not something educators can prevent. Avoiding engagement with AI creates its own set of risks, including leaving students unprepared for a technology-saturated world. What educators can do is prepare students for this changing landscape. AI offers significant educational potential and Vermont schools should explore these opportunities while teaching students critical thinking and ethical frameworks for responsible AI use.

Part 4: Implementation & School Leadership

Needs Assessment

Before developing or refining your district's approach to AI, assess your current state across these key areas. This will help identify gaps and prioritize next steps.

Policy & Governance

Many districts have found that their greatest needs related to AI policies are to update their existing policies with AI taken into consideration. For some districts/schools, these will exist in student handbooks and in others they are standalone policies. Both should be reviewed.

Have you updated your policies:

- Have your acceptable use policies been updated?
- Have your academic integrity policies been updated to include how students will know whether AI usage is allowed and to what extent? Examples of approaches can be found in the section on [academic honesty](#).
- Do bullying and harassment policies cover AI-generated content (deepfakes and NCII)? More information including a link to a sample policy is found in the section on [student well-being and safety](#).
- Are there any unified statements or policies around your district or school's approach to AI in education? Excellent examples include:

- Artificial Intelligence in CVSD – Principles and Beliefs³⁵
- Artificial Intelligence at EWSD includes their guiding principles for AI use³⁶
- MVSU Generative AI Guidelines³⁷
- Have your student and staff handbooks been reviewed with respect to AI?
- Has guidance with respect to protecting student data privacy been given to staff and educators?

Staff Readiness

Staff readiness varies significantly across schools and districts. This variation is an asset: early adopters can pilot approaches and share findings, while colleagues who move more cautiously provide time to develop training and surface important questions about AI's appropriate role. Before expecting educators to integrate AI into instruction, they need access to approved tools for their own use and opportunities to build skills through low-stakes practice. Teachers should gain considerable experience using AI before they guide their students' use of AI. [Principle #1 of AI usage in Education](#) focuses on administrative and planning tasks, and this provides ideal foundational experience. Assessing current comfort levels and training gaps helps prioritize professional development investments.

- What is the general comfort level of staff with AI tools?
- Have staff received any AI literacy training?
- Do staff have access to approved AI tools for their own use?
- Do educators have consistent messaging to provide students with respect to AI usage?
- Are there clear guidelines for when/how staff should use AI?

Student Learning

Student AI use should be guided by clear expectations and appropriate to students' developmental readiness. The [Developmental Appropriateness of AI](#) section provides a framework for scaffolding student access by grade band, while [Principle #3](#) outlines AI literacy competencies students need to develop.

- Is there a developmental framework for student AI use by grade level?

³⁵ [Artificial Intelligence in CVSD – Principles and Beliefs](#)

³⁶ [Artificial Intelligence at EWSD](#)

³⁷ [MVSU Generative AI Guidelines](#)

- Do students have access to any appropriate, vetted AI tools? Is this done on a per activity basis or do they always have access to these tools?
- Are expectations for AI use on assignments clearly communicated?
- Is there a plan for students to receive AI literacy instruction?

Infrastructure & Support

Successful AI implementation requires both approved tools and support systems to use these tools effectively. The [Tool Vetting Framework](#) provides guidance for evaluating and approving AI tools while ensuring compliance with data privacy requirements.

- Are approved AI tools available to staff who need them?
- Under what circumstances are students provided with AI tools?
- Are these tools approved by IT?
- Have you intentionally decided whether students should access to these tools at any time or only for specific activities?
- Are there ways to audit the usage of these AI tools?
- Are usage statistics informing decision-making?
- Are IT staff familiar with how to conduct investigations when required?
- Is there technical support for AI implementation?
- Is there a process for ongoing evaluation and adjustment?

Community Engagement

Community support is essential for successful AI integration. Families are partners in students' AI literacy development, while school board members and the broader community need to understand how AI is being used and why. Clear communication helps build trust and ensures decision-makers are informed. The [Parental & Guardians Guidance](#) section includes resources that can be shared with families.

- Have families been informed about the district's approach to AI?
- Are resources available to help families navigate AI at home?
- Is there a channel for families to ask questions or raise concerns about AI use in schools?
- Are communications about AI accessible to community members with varying levels of technical familiarity?
- Have school board members received information or training on AI in education?
- Is the community informed about AI investments and their educational rationale?

Tool Vetting Framework

In general, approving an AI tool for use in a school district is very similar to any other educational technology tool that can store FERPA protected data. These systems do introduce some unique challenges, however. The Future of Privacy Forum released a guide that goes into some of the unique challenges of these tools: [Vetting Generative AI Tools for Use in Schools](#). (Sallay, 2024)

Categories of AI tools

As of 2025, AI applications for education generally fall into one of three categories:

- General Purpose AI tools
- Education specific AI chatbots
- Education specific applications that incorporate AI

Important Note: The AOE does not recommend vendors. This section does mention products that are in use or likely to be in use in Vermont schools.

General Purpose AI Tools

This category includes generative AI tools that were built for a general audience and are not specific to education. This includes chatbots like ChatGPT, Claude and Gemini, but it also includes image, video, and audio generation tools like Dall-E, Stable Diffusion and Otter AI. Some of these tools are available in education specific versions, but their feature sets are similar.

Generic Chatbots

Google has made Gemini and NotebookLM free for education, and as of the writing of this document all Vermont schools are using the Google for Education platform. Google trained an LLM AI specifically for education, LearnLM³⁸, and their AI is covered by their educational privacy policy (when using school accounts only) which gives it an advantage over other generic chatbots. Gemini is also baked into Google Classroom and Google Docs, integrating it with tools that students and teachers regularly utilize.

Google is providing some auditing features with Gemini. Unfortunately, as of this document's release, they are quite limited and require a Google Administrator. This means that teachers are unable to have full access to all of the conversations that their students have with Gemini. Even for a Google Admin, most of the access is through Google Vault and only applies to the standalone Gemini app and not where the app is embedded into other Google tools. IT departments need to actively set the conversation

³⁸ [LearnLM - Google](#)

retention settings and teachers need to clearly understand which conversation should not occur in Gemini in apps because of the lack of retention/logging.

Generic Chatbots for Teachers

Providing teachers with access to an official AI tool can mitigate the risks of teachers using public chatbots with student protected data.

Access to a generic chatbot like Gemini does require teacher training, including discussions of when the use of AI is appropriate.

Being open in nature, teachers need to develop skills with prompting to fully leverage the AI and a prompt library can be helpful as a training and resource. Google is introducing the ability to share GEMs and notebooks and creating these using predeveloped prompts and resources can be a great way to empower educators. For example, create a GEM that contains curricular resources as well as curricular standards and supply an initial prompt (take a look at Ethan Mollick’s “Lesson Crafter” prompt³⁹, AI for Education’s Prompt Library⁴⁰ or Microsoft’s prompts for educators⁴¹ for ideas).

Generic Chatbots for Students

It is impossible to fully control the use cases for generic chatbots. This can easily lead to students bypassing the learning process or for them to view their relationship with the AI as more than the relationship with a technological tool. AOE’s recommendation is that students in K-8 not have access to these tools (See [Developmental Appropriateness of AI](#)). In a high school setting schools should pilot these tools first paired with an AI literacy strategy, before considering whole school rollouts.

Audio and Meeting-based AI

There are some powerful use cases for AI audio tools that cover transcription, note taking, and translation. There are a huge number of apps in this category, including ones that are built into the major webinar platforms. These can be powerful tools for:

- Taking meeting notes
- Generating class notes
- Enabling near real-time translation services for supporting ELL students and families
- Supporting teacher coaching in lieu of classroom observations

³⁹ [Instructor Prompt – Ethan Mollick](#)

⁴⁰ [GenAI Chatbot Prompt Library for Educators – AI for Education](#)

⁴¹ [Prompts for Education - Microsoft](#)

Some of the concerns include:

- Privacy being violated through the use of insecure systems or oversharing of generated notes. FERPA and HIPPA (medical data) rules need to be enforced. Because it is impossible to control the content of a conversation, vetted tools must assume the highest level of privacy requirements.
- Students recording class and sharing generated transcriptions or notes in inappropriate ways.
- Participants feeling less comfortable being open and honest
- Recordings made secretly without the awareness of all participants

Actions to take include:

- Ensuring that schools have policies/procedures in place govern recordings made in class by both students and staff
- Providing staff with vetted tools to use if recordings are allowed

Resources:

- [Classroom Recordings and FERPA: What You Need to Know | GovFacts](#)

Education Specific AI Chatbots

A large number of new EdTech companies have launched AI solutions for education. These tools leverage existing AI models (ChatGPT, Claude, Gemini) behind the scenes but provide a structured user interface that is easier for teachers to get started in and provides more tightly structured student experiences. Popular products in this space include: [Brisk Teaching](#), [MagicSchool AI](#), [Khanmigo](#), and [SchoolAI](#).

Most of these tools support both an educator-only mode and a student-facing mode. They tend to need less prompting experience on the part of the user, since they have prebuilt tools rather than expecting the user to provide context to the AI system. This can limit these tools but can also make them more accessible to a wider audience with limited training.

Education Specific AI Chatbots for Students

When evaluating the student-facing aspect of these tools, it is important to learn if students can interact with the system at any time (tutoring features sometimes work this way) or if students can only use the tools under strict structures put in place by the teacher. Consider what management tools the teacher has access to and whether they can review the conversations that the students engage in.

There are concerns about the risks vs. benefits of exposing young students to these tools. More open access to the tools should follow AI literacy lessons where students learn about these technologies and how to leverage them for learning. Please consult the section on the [developmental appropriateness of AI](#).

Education Specific Applications That Incorporate AI

Several educational applications are incorporating AI features as a component of their product. The use of AI in these tools looks less like the generic chatbot and instead serves as a focused support to the larger product. Examples include:

- Literacy screening tools using AI to assess students based on students reading aloud and explaining passages of text.
- Curricular materials that incorporate AI tutoring but only within the narrow focus of the content.

Parental and Guardians Guidance

Parents and guardians can be supportive allies of schools, supporting students' development of their AI literacy skills. Clear communication, learning opportunities and resources should align with the AI strategy that your district has developed.

A number of organizations have released parental guidance. Consider aligning these resources with your district's developmental timeline for student use of AI.

Specific Resources you can share with parents and guardians include:

- [Parents' Ultimate Guide to Generative AI - Common Sense Media](#)
- [Digital Parenting Resources - Family Online Safety Institute](#)
- [AI Companion Apps - Parent Tool Kit from Cyber Safety Project](#)
- [A Quick Guide to Misinformation & Media Literacy - ConnectSafely](#)

Part 5: Resources & Appendices

Frequently Asked Questions

What do you mean by AI?

Artificial intelligence (AI) is technology that enables computers to perform tasks normally requiring human-reasoning, decision making and creativity. While different forms of AI have been in use since the 1970s, generative AI systems like ChatGPT have only been available since late 2022.

This document uses "AI" to refer broadly to these systems but focuses primarily on generative AI tools - particularly Large Language Models like ChatGPT, Claude, and Gemini that can understand and create text, as well as multimodal systems that work with images, audio, and video.

What if I don't know anything about AI?

You are not alone. AI systems like ChatGPT have only been widely available since late 2022. The next question in this FAQ will point you toward professional development resources that you can use.

One of the best ways to learn about these systems is to explore them within your area of expertise. When you understand a topic deeply, you can better evaluate AI's responses, catch errors, and guide the conversation productively—skills that matter whether you're a new user or an experienced one. Try using an AI chatbot like Gemini or ChatGPT with a prompt grounded in your current work:

"I teach [subject] to [grade level] students and I'm working on [specific lesson, unit, or challenge]. Before making suggestions, ask me 2-3 questions so you can better understand what I need."

How can I access training and professional development?

The AOE will be releasing new professional development and sharing information on high quality training materials as well.

Online training resources can be found in the [Professional Development Resources](#) section.

Which AI tools can I use in my classroom?

While this guide explores the opportunities, it does not try to definitively answer whether teachers are allowed or should use AI in a specific situation. Instead, this guide clearly indicates the points of caution and considerations for each situation.

Most supervisory unions and districts maintain an approved list of software that can be used in the classroom and that store student data. FERPA, COPPA and other applicable laws govern how student data must be handled. Most districts have signed data privacy agreements with specific vendors and require that teachers utilize these vendors platforms. You should reach out to your district technology team to determine which AI tools you may use. This guide includes information on [how to properly vet and approve AI applications for use in districts](#).

In some cases, the question is not what AI tools you can use, but how you can use them. These discussions should occur in your school, with decisions made by school leadership. For instance, before using AI as an assessment tool to provide feedback on students' work, you should consult with other educators within your school and seek the approval of your school leadership.

What if my students use AI inappropriately and /or without permission?

Students using AI without permission is a concern, but it is manageable with clear expectations, good pedagogy and clear school wide standards and policy. When students use AI without permission, they can cheat themselves out of learning the material and potentially violate the school's academic honesty policies. In principles #3 and #4, this guide will provide advice for teaching AI literacy and designing assignments that expect a certain amount of student AI usage.

In addition to the need for schools to update policies, it is also important that teachers be clear about expectations with students. AI tools are being baked into the tools that we use every day, being embedded in Microsoft Word, Google Docs and other tools that are frequently available to students. A grammar check utilizes a form of AI, and a tool like Grammarly goes beyond a traditional grammar check to make more robust editorial suggestions. Students may use AI for research, brainstorming ideas, improving their writing style, or giving them feedback on early drafts. Students may not be clear about what constitutes AI usage and what is allowed or not. They may also not be clear about when a citation is required. It is important that you are clear about your expectations in your class and on assignments.

There are tools available that claim to detect AI generated text. They work by using statistical analysis and can work in some simple cases. Unfortunately, they can have false positives and are not a reliable method of identifying academic honesty violations. In almost every case, the teacher will be a better judge than an automated AI detection tool.

Please read the [Academic Honesty](#) section under risk mitigation strategies for more info.

How can I minimize academic honesty violations related to student use of AI?

Academic dishonesty was common before the introduction of AI and it continues to be common after its introduction. An ongoing research study by Stanford researchers suggested that cheating amongst U.S. High Schoolers did not increase with the release of ChatGPT (Spector, 2023). Students who are academically dishonest are overly focused on the final product and not the learning process. As teachers we need to refocus our students on the learning process. Ensure that your assessments evaluate this process in addition to the final product.

For instance, for a writing assignment, you can focus on the early drafts and meet individually with students to determine what they intend to improve. Treat these meetings as a formative assessment, asking the students why they made the choices they made and what they think can be improved.

It is important to include some in class assessment techniques, and as demonstrated above, these do not have to be traditional tests. By using a variety of assessments and creating lots of opportunities for conversations with students, you can provide students with feedback and support that can reduce the desire to cheat.

Please read the [Academic Honesty](#) section under risk mitigation strategies for more info.

How to Work with AI for Educators (Prompting Guide)

A few tips and techniques can assist when utilizing AI.

Stay in the driver's seat

Read and edit everything the AI generates. You will get better output by remaining in charge of the entire process. The AI does not have the full context and does not have the knowledge or good judgement that you have, which can lead to “hallucinations” and mistakes. Take the time to adjust even the style and tone of the AI's outputs to be your own.

The new AI tools are excellent writers and sound confident even when they are making something up. Read the output carefully before you use it, it is very easy to get complacent and overly trust AI output.

Be transparent

As you utilize AI, share with your colleagues and students how you are using it. Students in universities have complained when their professors utilized AI extensively but explicitly didn't allow students to utilize AI (Hill, 2025). Being transparent demonstrates the same AI Literacy skills that you will want to help your students learn. Consider telling your students that you are experimenting with AI, before assigning any use of AI to them, so that you can learn the best ways to help your students use these tools responsibly.

Of note, both MLA and APA support citing AI generated content⁴². Make a point of indicating AI usage in both your documents as well as clearly indicating AI generated images in any presentation that you create.

Take an incremental approach

Start with low-stakes administrative tasks like editing emails and brainstorming activities before moving to lesson planning or curriculum development. Build your skills and confidence gradually, learning to recognize AI's strengths and limitations through experience with less critical tasks first.

⁴² [Harvard's AI Citation Guide](#)

Provide proper context

The AI does not know who you are or anything about your school, grade level you teach, your students' interests, your current lesson, or even what country you are in! It needs to generate its response based solely on the context it is provided. The clearer the context you provide the better the output will be. Even the tone of the AI's writing is completely variable, with the AI trying to match the context it is provided (typically based on what you type for a prompt). If you can, provide the AI with documents to help provide greater context. For instance, if you are asking the AI to help develop assignments based on a reading you are asking the students to do, upload the reading to the AI.

Try again and have a conversation

Sometimes the best output is generated through a conversation and not just a well-developed single prompt. Try telling the AI to ask you clarifying questions first, before it tries to solve your request. This way it can better understand your needs before making any suggestions.

If the conversation goes in the wrong direction, restart the session instead of trying to steer it in the correct direction. It is a waste of time and effort to try to correct the AI. In most AI systems, the AI does not “remember” the conversations that you have had with it, so you will need to provide the full context each time. In a system like Gemini or ChatGPT you can edit your prompt from previously in the chat to restart the conversation from an earlier point instead of starting from scratch.

Give the AI an example

In AI lingo, giving the AI an example of good work is called “single shot prompting” and multiple examples is called “few shot prompting”. Example: If you want the AI to generate a lesson plan for you, supply a sample of a high-quality lesson plan so that the AI can better understand your expectations.

Build a custom AI

Most AI systems like Gemini and NotebookLM allow you to prebuild an AI chat, providing reference documents and information. You can then reuse and even share this chatbot with other people. Consider a chatbot preloaded with curricular standards and curricular materials for your grade level that can serve multiple teachers. Then anytime you need to develop curriculum, you don't need to reload the reference documents for the chatbot to have the proper context.

Professional Development Resources

Consider what foundational professional development educators should receive that align with your AI strategy.

Notable free foundational training courses include:

- [ISTE+ASCD: 5-hour self-paced and 15-hour facilitated courses as well as other PD opportunities. Scholarships opportunities are available](#)
- [AI for Education – Khan Academy](#) – Includes content from ISTE, Code.org and ETS
- [AI Literacy for Educators course](#) from Massachusetts DESE
- [Empowering Educators in the Age of AI | NEA](#)

Educators looking to better understand how AI works may want to explore:

- [How Generative AI works](#)
- [Free The AI Revolution Tutorial at GCFTGlobal](#)

Glossary of Terms

A

- **Agentic AI Systems:** AI tools capable of acting autonomously or semi-autonomously to complete multi-step tasks toward a goal with limited human input.
- **AI Companion:** A chatbot or digital avatar designed for social interaction, raising concerns about emotional safety and dependency in younger users.
- **AI Fluency:** The ability to confidently and appropriately use AI tools, including understanding limitations, ethical implications, and best practices.
- **AI Literacy:** Skills needed to understand what AI is, how it works, when to use it appropriately, and how to evaluate its output critically.
- **Algorithm:** A set of step-by-step instructions that allows computers to analyze data and accomplish tasks.
- **Alignment:** The process of adjusting AI systems to produce desired outcomes and adhere to human values or goals.
- **Anthropomorphism:** Attributing human-like traits or emotions to AI systems even though they lack consciousness.
- **Artificial Intelligence:** Computer systems designed to perform tasks that typically require human intelligence, such as understanding language, recognizing patterns, or making predictions.

B

- **Bias:** Systematic errors in AI outputs caused by skewed or unrepresentative training data.

C

- **Cognitive Offloading:** Using AI or technology to handle cognitive tasks. Helpful at times but harmful when it interferes with learning.
- **Context Window:** The input to a large language AI model.

D

- **Dataset:** A structured collection of data used to train, test, and evaluate AI models.
- **Emergent Behavior:** Unexpected abilities exhibited by large AI models as they scale.

F

- **Few-Shot Prompting:** Providing several examples to help the AI generate more accurate or relevant output.

G

- **Generative AI:** AI systems that create new text, images, code, or media by learning patterns from data.

H

- **Hallucination:** When AI generates plausible but factually incorrect information.

L

- **Large Language Model (LLM):** A powerful model trained on massive text data able to generate language.
- **Machine Learning (ML):** AI subset where systems learn patterns and improve from data without being explicitly programmed.
- **Multimodal Model:** An AI model that processes and generates multiple types of data (text, images, audio).

P

- **Prompt Engineering:** Crafting prompts to guide AI toward better or specific outputs.

T

- **Token:** The basic unit of text (parts of words) that AI models process.

Quick Start Guides

Quick Start Guide: Elementary Educators (PreK-5)

Priority Focus

Elementary educators should concentrate on ways to use AI to streamline their own work.

Professional Practice

- [Principle #1: Streamline Administrative Work](#) - Use AI to save time on lesson planning, differentiation, and communication
- [Principle #2: Use AI to support personalized learning](#) - focus on ways the educators can generate personalized materials and provide better accommodations to meet individualized student needs.
- [How to Work with AI for Educators\(Prompting Guide\)](#) - Learn basic techniques to get better results

Supporting Student Use of AI

- PreK-2: This guide recommends students NOT use AI directly at this age. You may use AI-powered tools for instruction. *Check with your school leadership about your district's approach.*
- Grades 3-5: [Review developmental guidelines](#) - This guide suggests students only use AI embedded in curriculum-specific applications at this stage. *Your district may have different guidelines.*
- [Principle #3: Develop Student AI Literacy](#) - Teach AI awareness appropriate for your students' readiness level

Understanding the Risks

- [Part 3: Responsible AI Usage](#) - Learn about data privacy, bias, and when NOT to use AI

Getting Started

- [Professional Development Resources](#) - Find training opportunities

Quick Start Guide: Middle School Educators (6-8)

Priority Focus

Middle school is a transition point, where students need to develop AI literacy skills with structured guidance.

Professional Practice

- [Principle #1: Streamline Administrative Work](#) - Use AI to save time on lesson planning, differentiation, and communication
- [Principle #2: Use AI to support personalized learning](#) - Explore AI tutoring and adaptive learning tools
- [How to Work with AI for Educators \(Prompting Guide\)](#) - Improve your prompting skills

Supporting Student Use of AI

- [Developmental Guidelines \(Grades 6-8\)](#) - This guide recommends developing students AI literacy skills and having students use education-specific chatbots under teacher direction. *Your district may have different guidelines.*
- [Principle #3: Develop Student AI Literacy](#) - Build structured AI literacy curriculum tailored to your students' readiness
- [Academic Honesty Strategies](#) - Establish clear expectations for AI use in your classroom

Understanding the Risks

- [Part 3: Responsible AI Usage](#) - Understand risks including cognitive offloading and mental health concerns

Getting Started

- [Professional Development Resources](#) - Find training opportunities

Quick Start Guide: High School Educators (9-12)

Priority Focus

High school students increasingly need AI fluency to prepare for college and careers in an AI-integrated world.

Professional Practice

- [Principle #1: Streamline Administrative Work](#) - Use AI to save time on lesson planning, differentiation, and communication
- [Principle #2: Use AI to support personalized learning](#) - Explore AI tutoring and ways to meet student accommodations
- [How to Work with AI for Educators \(Prompting Guide\)](#) - Improve your prompting skills

Supporting Student Use of AI

- [Developmental Guidelines \(Grades 9-12\)](#) - This guide suggests students may be ready to use AI for more cognitively demanding tasks once they have developed their AI Literacy skills. *Consult your school's policies and consider your students' readiness.*
- [Principle #3: Develop Student AI Literacy](#) - Build structured AI literacy curriculum tailored to your students' readiness
- [Principle #4: AI-Infused Cognitively Demanding Tasks](#) - Design assignments where AI encourages higher-order thinking by students
- [Academic Honesty Strategies](#) - Establish clear expectations for AI use in your classroom

Essential Reading

- [Part 3: Responsible AI Usage](#) - Understand risks including cognitive offloading and mental health concerns
- [Ethics of AI Use](#) - Understands the impacts AI may have on the world so that you can better prepare students

Getting Started

- [Professional Development Resources](#) - Find training opportunities

Quick Start Guide: IT & School Leaders

Priority Focus

Establishing infrastructure, policy, professional development and support systems for safe AI integration.

Strategic Planning

- [Needs Assessment](#) – Begin with a needs assessment to identify where to focus your efforts

Technical Requirements

- [Tool Vetting Framework](#) - Establish procedures for evaluating and approving AI tools
- [Data Privacy](#) - Ensure FERPA, COPPA, and HIPAA compliance

Policy Development

- [Policy and Procedure Development](#) - The needs assessment includes information for updating policies and handbooks
- [Risk Mitigation Strategies](#) - Learn about ways to safeguard academic honesty, student mental health, student learning, student wellbeing and safety

Student Learning

- [Developmental Appropriateness of AI](#) for scaffolding AI Literacy and student AI usage
- [The Four Principles Framework](#) for understanding use cases for AI in education

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